

Prototyping

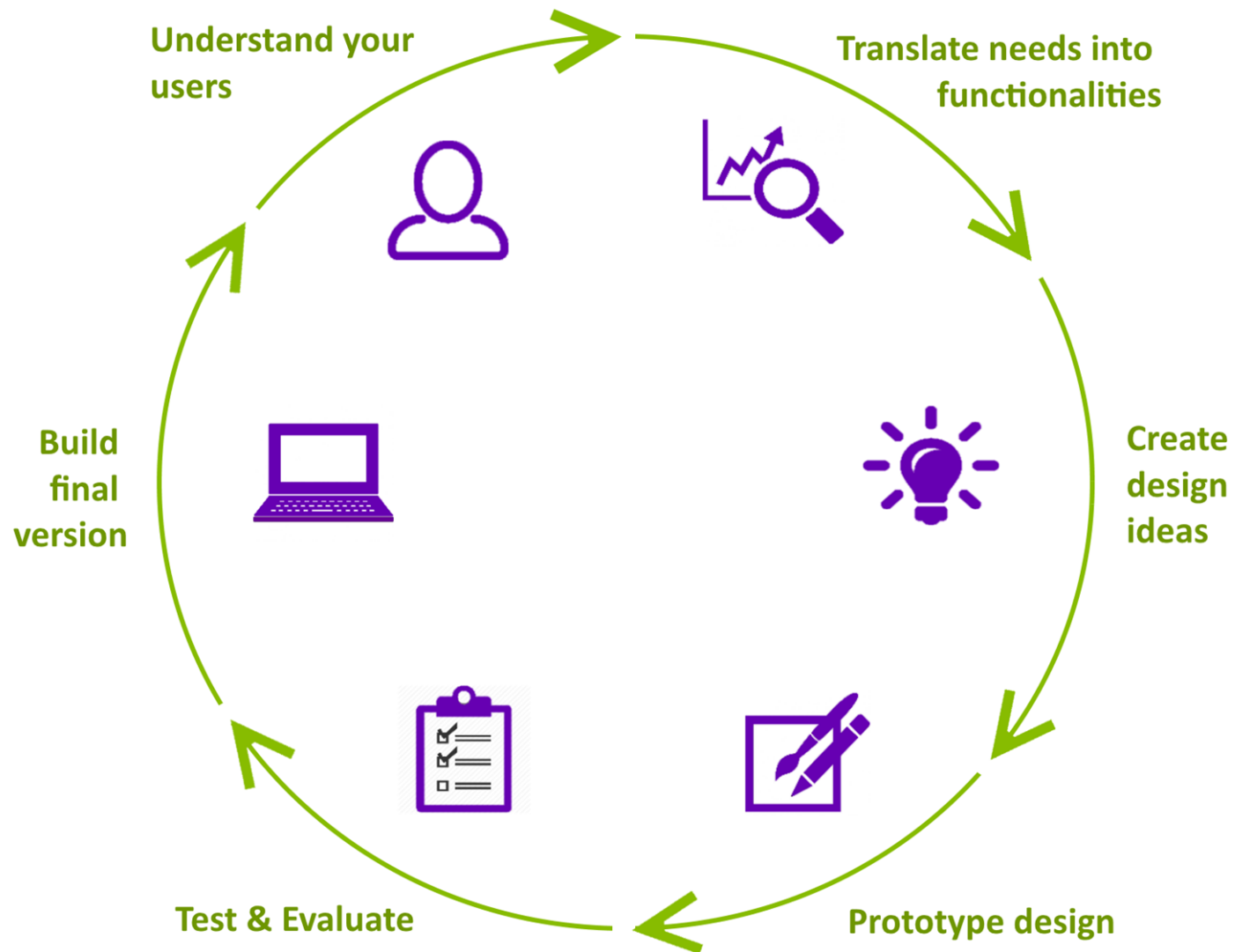
Low-Fidelity Prototyping

High-Fidelity Prototyping

Prototyping Dimensions

Announcement

- Lab 0 marks: released
- Lab 1 marks: released
- Lab 2 marks: TAs are working on
- Project step 1: due this Friday (October 11 at 11:59 pm)



IDEO Shopping Cart

- <https://www.youtube.com/watch?v=M66ZU2PClCM>



What is a Prototype?

- In other design fields a prototype is a small-scale model:
 - a miniature car
 - a miniature building or town
 - a juicer



What is a Prototype?

- In interaction design it can be (among other things):
 - a series of screen sketches, i.e. not necessarily related
 - a storyboard, i.e. a cartoon-like series of scenes
 - a Powerpoint slide show, e.g., simulation
 - a video simulating the use of a system
 - a cardboard mock-up
 - a piece of software with limited functionality written in a language
- Key point:
 - Goal is to NOT the artifact, but the feedback

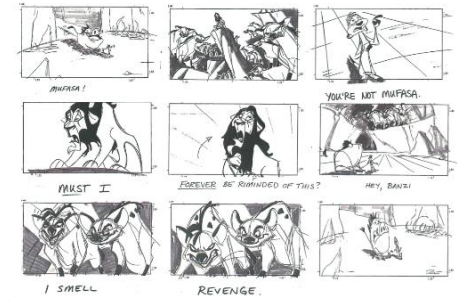


Image source: Nintendo

Sketch of A Screen

HTML Page

Student Information Help

Student Number: 789-567-234

First Name:

Middle:

Surname:

Salutation:

Date first Enrolled: June 14 2003

Seminars:

Seminar	Term	Mark	Status
CSC 100 Intro to CS	Fall 2003	A+	Passed
CSC 200 Intro to AM	Fall 2003	A	Passed
CSC 203 Advanced AM	Spring 2004	-	Enrolled

Edit Student Information - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Back Forward Stop Home Search Favorites Media Links

Student number: 789-456-123

First name:

Middle name:

Surname:

Salutation:

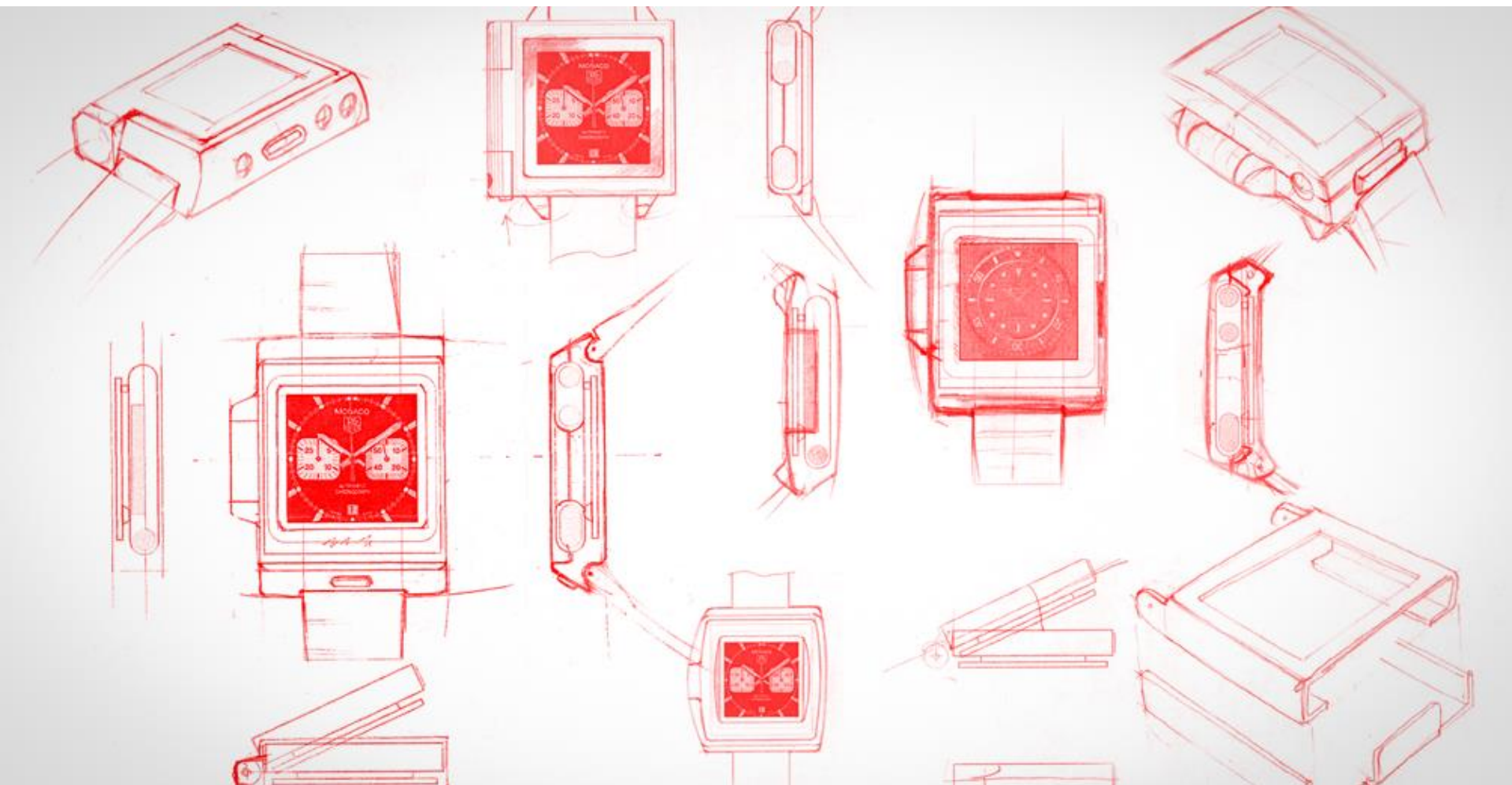
First enrolled: June 14 2003

Schedule:

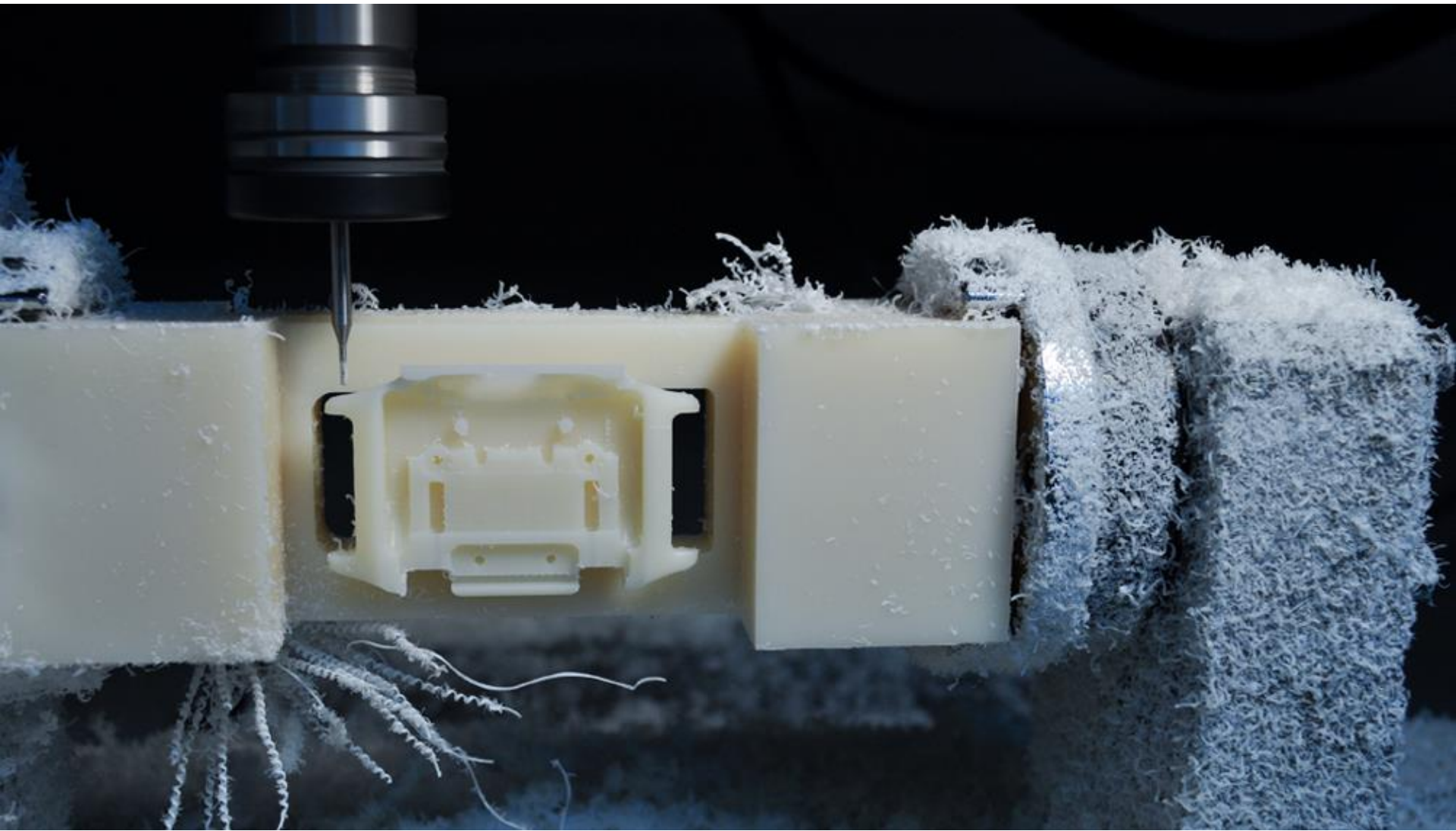
Seminar	Term	Mark	Status
CSC 100 Intro to C#	Fall 2003	A+	Passed
CSC 200 Intro to Agile Modeling	Fall 2003	B-	Passed
CSC 203 Advanced Agile Modeling	Spring 2004	-	Enrolled
CSC 220 Intro to Agile Databases	Spring 2004	-	Enrolled

Sketching

- Simulate a user experience with enough fidelity to allow third-party comment on the value of the new product.



3D Model



Source: <https://www.design1st.com/how-much-do-hw-prototypes-cost/>

Source: <https://www.design1st.com/how-much-do-hw-prototypes-cost/>

High-Fidelity Prototype



Source: <https://www.design1st.com/how-much-do-hw-prototypes-cost/>

Example: Kodak DC210 Prototype

- A digital camera developed at IDEO, at a time when such Cameras had not yet broken into the consumer space.
- IDEO was tasked to find a way to make sense of possible digital camera interactions



Why Prototype?

- **Communication:** discuss ideas with stakeholders
 - “Where’s the ON button?”
- **Evaluate** interface effectiveness for communicating conceptual model
 - “Whoops, users didn’t understand that.”
- Develop **requirements** and/or specifications •
 - “Uh-oh, here’s something we forgot.”
- **Learning** and problem solving
 - “Hey, that will work!”
- Further develop **conceptual and physical design**
 - “That’s way too heavy”
- **Save time and money**
 - Don’t waste time coding/building the wrong thing

What to Prototype?

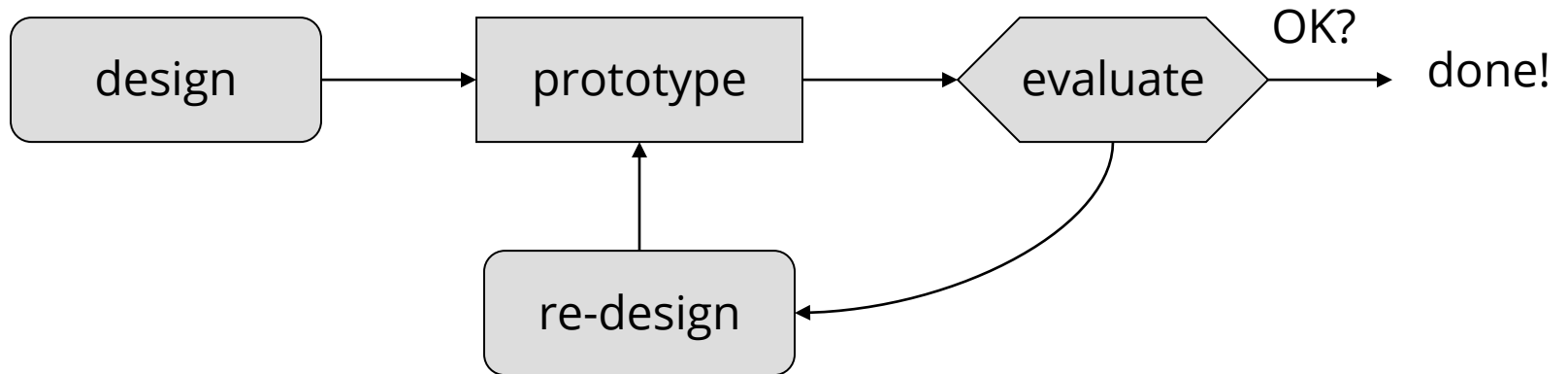
- **Screen layouts and information display:** placement of different widgets, types of widgets, etc
 - e.g. should edit fields come after/before control buttons
- **Work flow, task design:** the sequence of steps required to achieve the task
 - e.g. flow from one screen to another, changes on the screen
- **Difficult, controversial, critical areas:** provide alternative solutions to be assessed
 - e.g. business critical reports with specific alignment of fields
- **Technical issues:** will the graphics rendered be fast enough?

When to prototype?

- To get out of an **early**, focused discussion, and to reach agreement
- when you need to **communicate ideas**
 - design team, managers, users etc.
- when you have questions and you can't proceed:
 - **functionality:**
structure, sequencing, flow, clarity & completeness of information
Example: knob versus slider for controlling volume
 - **appearance:**
branding, clarity, aesthetics, color, shape, etc.
screen too crowded? actions clear, or lost in clutter?
 - **specifications:**
“design by prototyping” (evolutionary approach)
how many files to show in file selection box

Prototyping Process

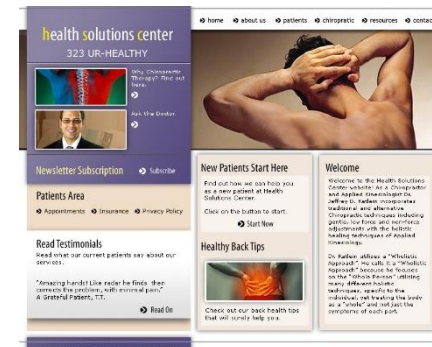
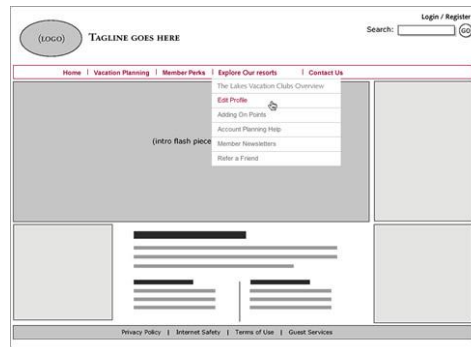
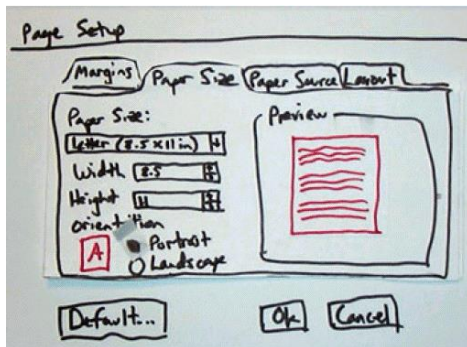
- You never get it right first time
- If at first you don't succeed ...



Prototype Fidelity

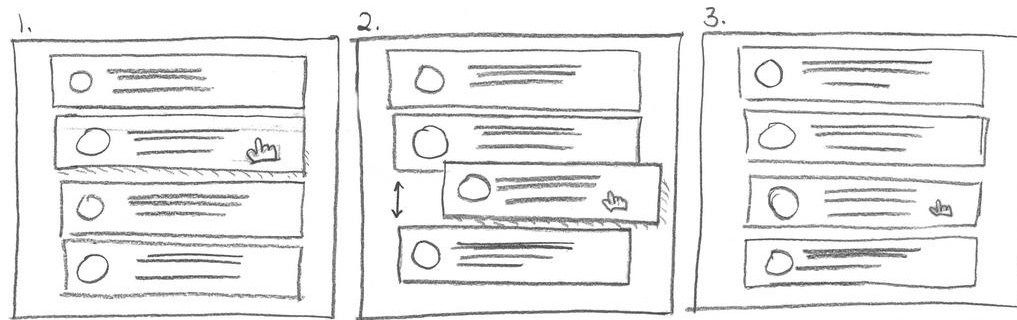
There are many types of prototypes, ranging anywhere between these two extremes:

- **Low fidelity:** prototype doesn't look much like final product, operation of the prototype may be simulated and slower.
- **High fidelity:** prototype looks and operates like the real product.



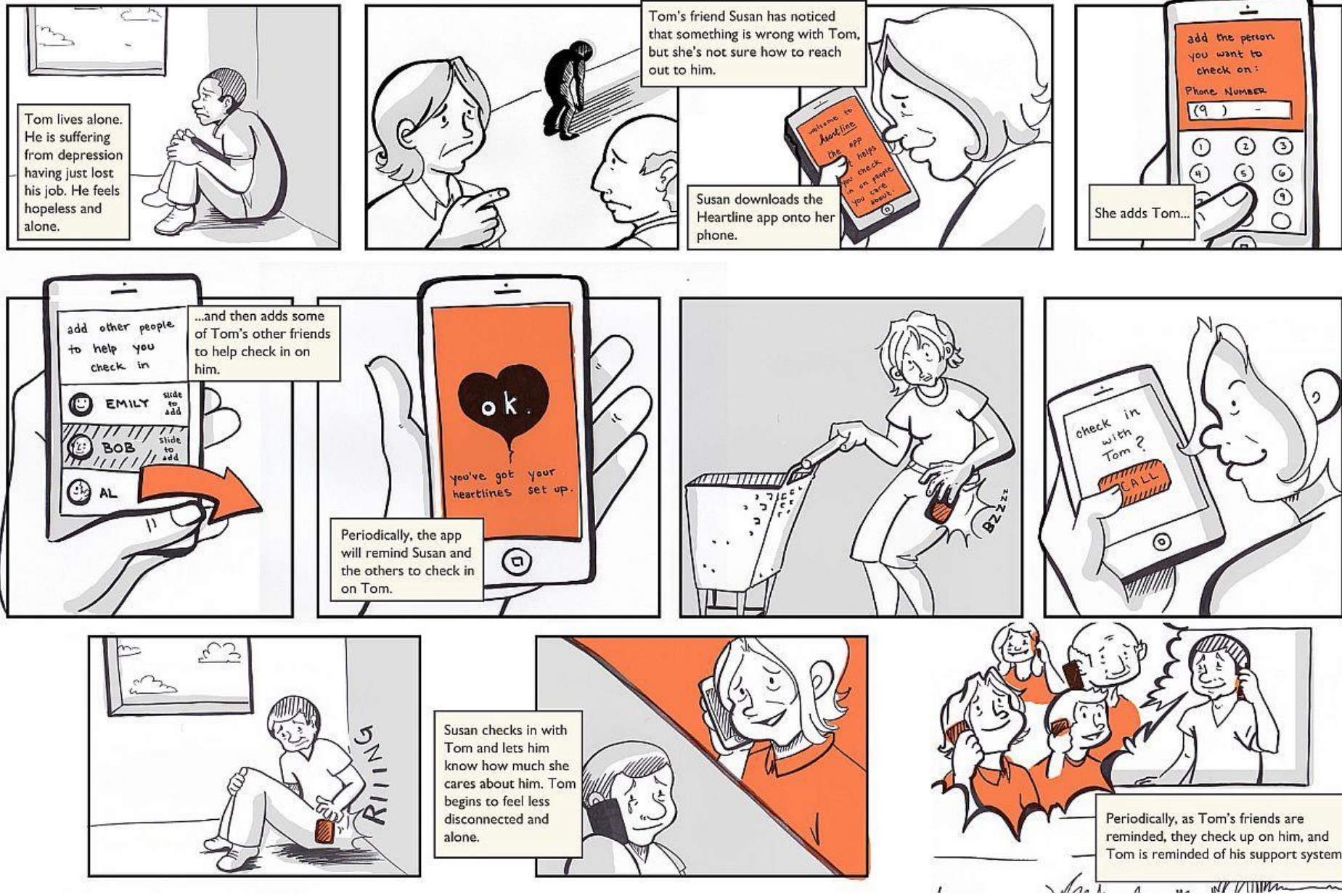
Low-Fidelity Prototyping

- Low-fidelity prototypes use a medium which is unlike the final medium, e.g. paper, cardboard



- Purpose:
 - Brainstorm competing representations
 - Elicit user reactions / suggestions
- Examples:
 - Storyboards
 - Paper prototypes
 - Wireframes

Storyboards Example: A Reminder Software



Storyboarding

- All about user tasks, not interface
- Like comic strip with bad drawing
- Conveys setting:
 - People
 - Environment
 - Task being accomplished
- Conveys sequence:
 - Steps involved
 - What task is being illustrated?
- Conveys satisfaction:
 - What motivates people to use this system?

Example: No Cell Phone Software

Uses
"star"
people



Benefits of Storyboarding

- Holistic focus: helps emphasize how an interface accomplishes a task
- Avoid commitment to a particular user interface
- Helps get all the stockholders on the same page in terms of their goal

Paper Prototypes

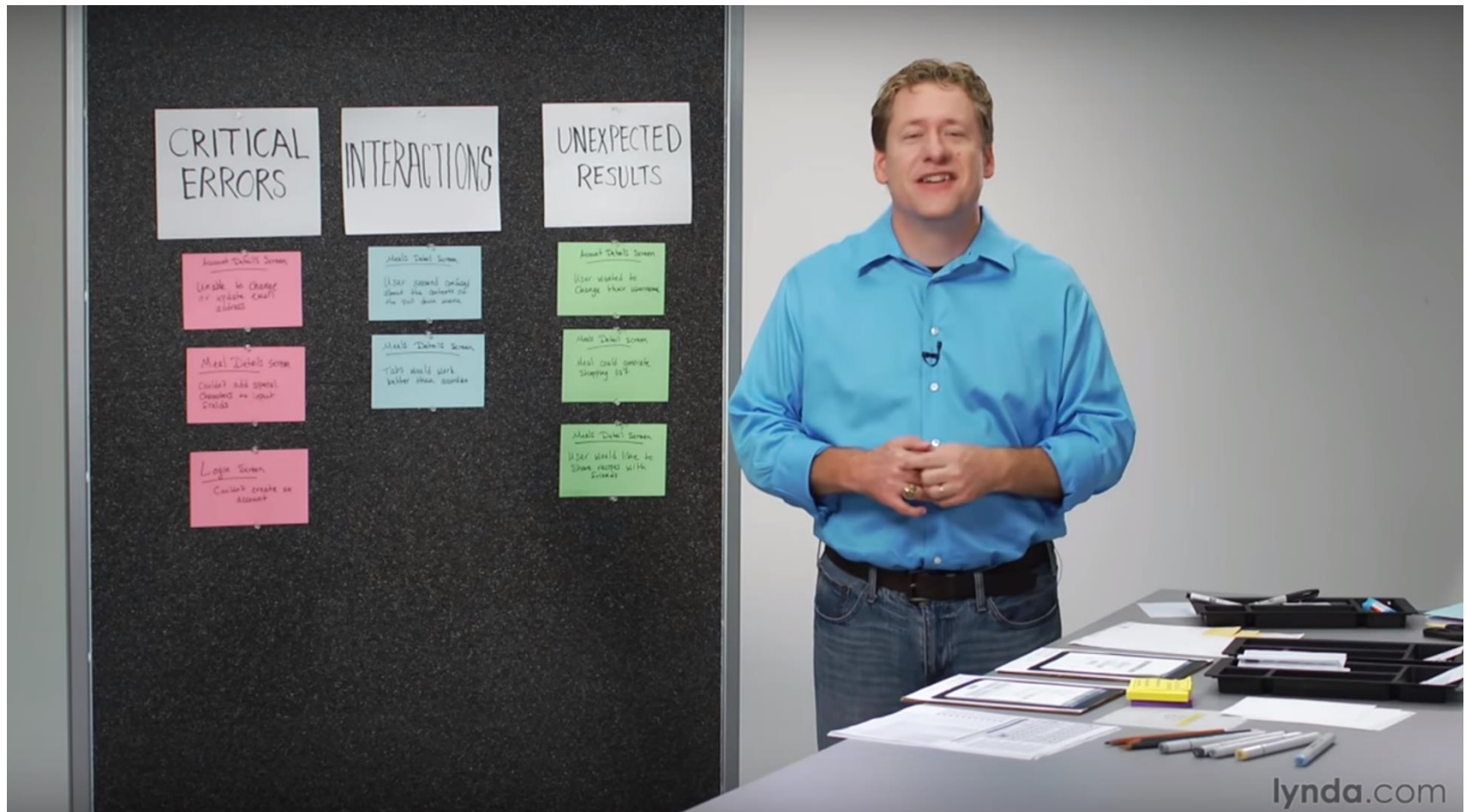
- Paper Prototypes
 - After storyboarding – know what your app accomplishes
 - Best for figuring out the user interface (UI)
 - Don't have to look beautiful or exact



Image source: UX Playground

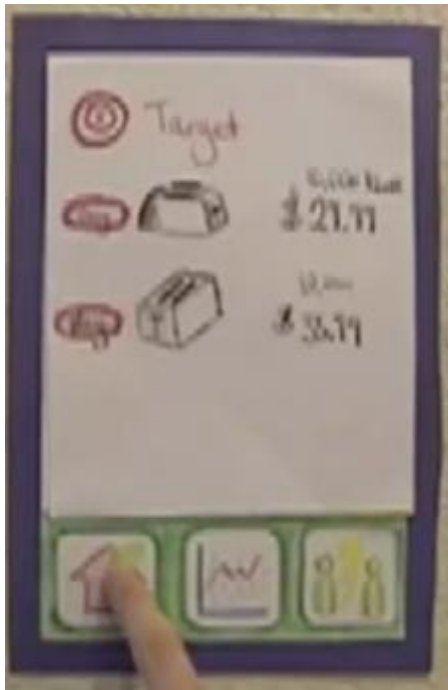
Paper Prototypes

<http://www.youtube.com/watch?v=FS00UIo12Xk>



Paper Prototypes

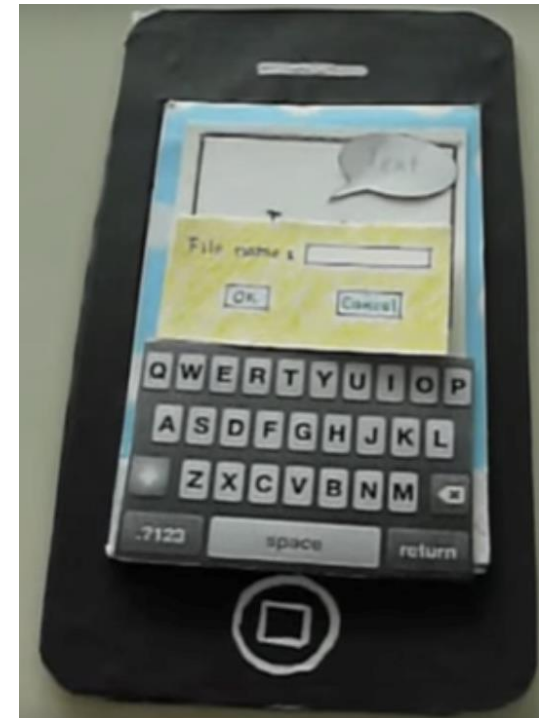
Show Interface Layout



Show Menu Items



Show Popups

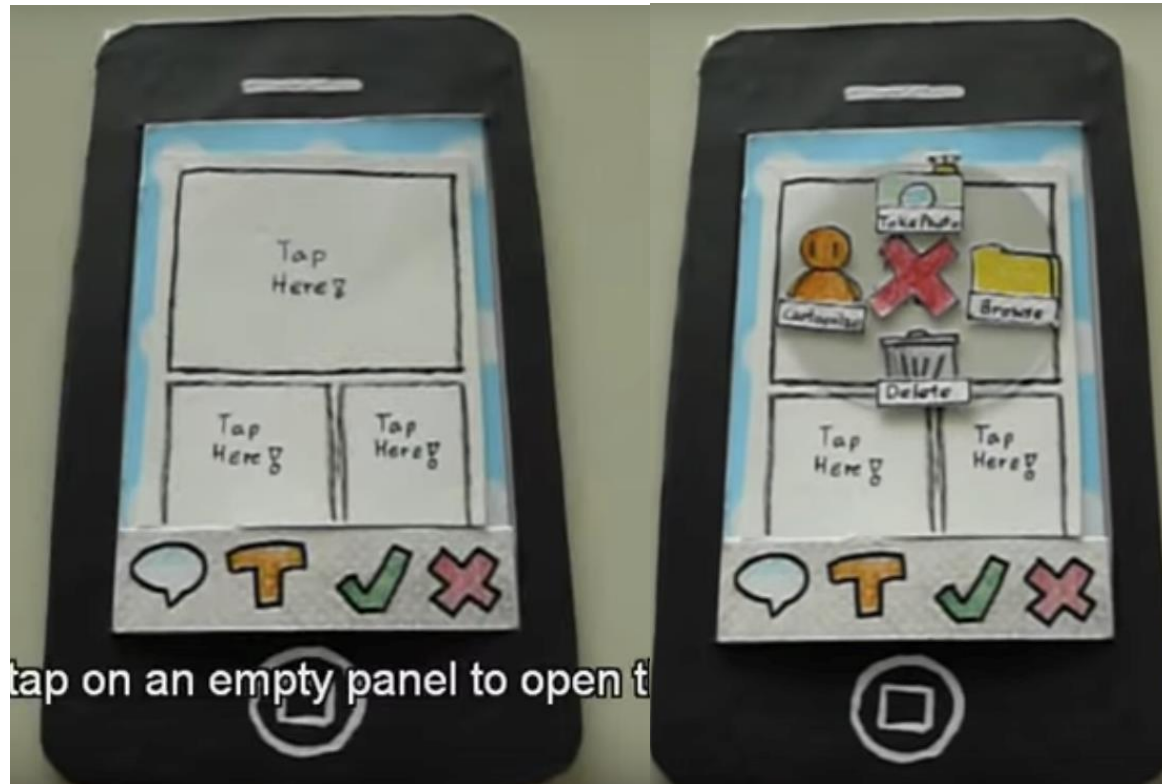


Paper Prototypes

Show Loading Time



Show Selection Changes



Show Sliding Window



Paper Prototyping

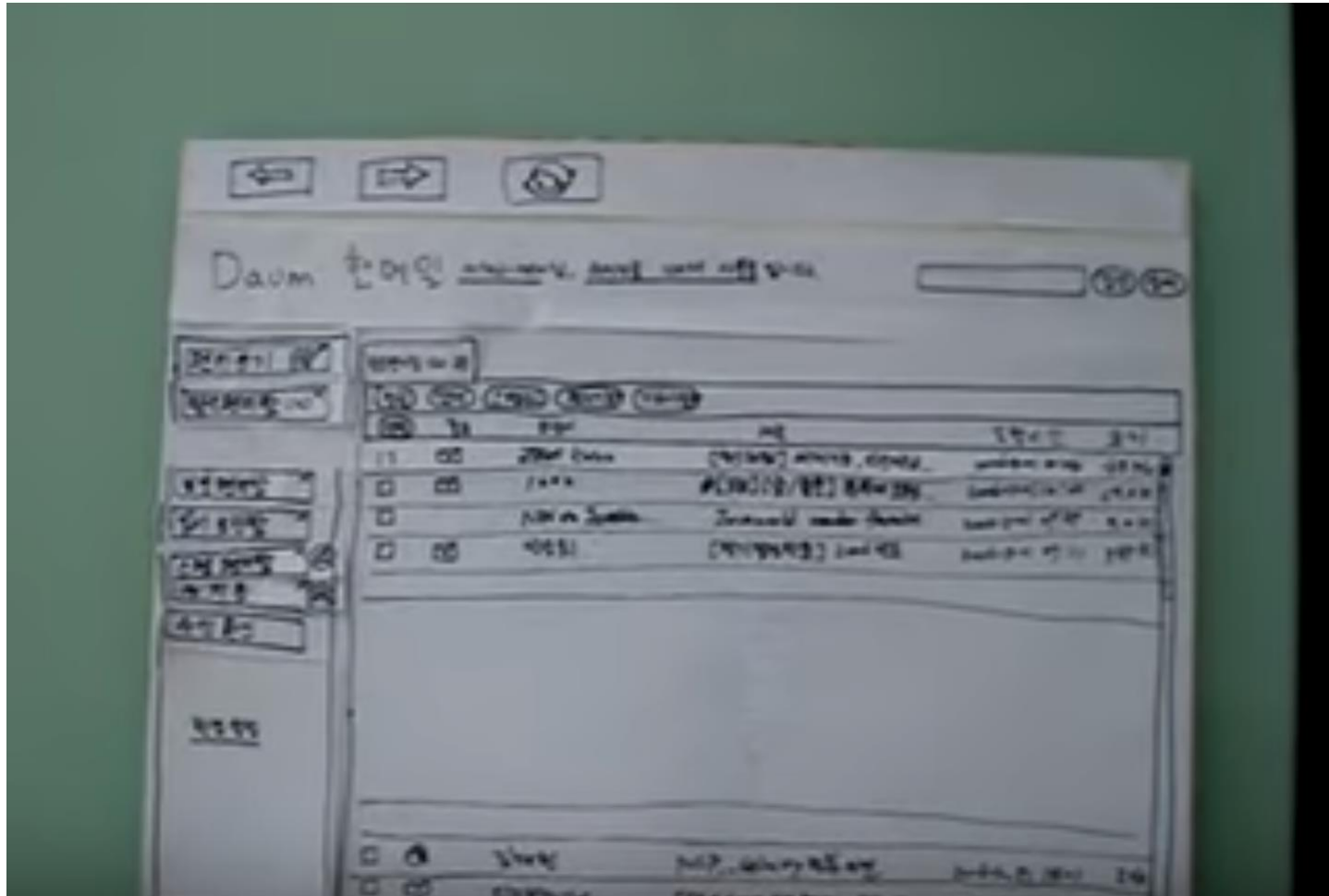
- Benefits
 - **Leverage common design skills:** Everyone can sketch and can build paper prototypes.
 - **Allow early testing:** testing prototypes early lets product teams find big-picture problems
 - **Support rapid experimentation:** Different user interface elements can be drawn, cut out, copied to make extras, and then assembled on a new piece of paper
 - **Facilitate adjustments:** easy to make changes during the testing session

Paper Prototyping

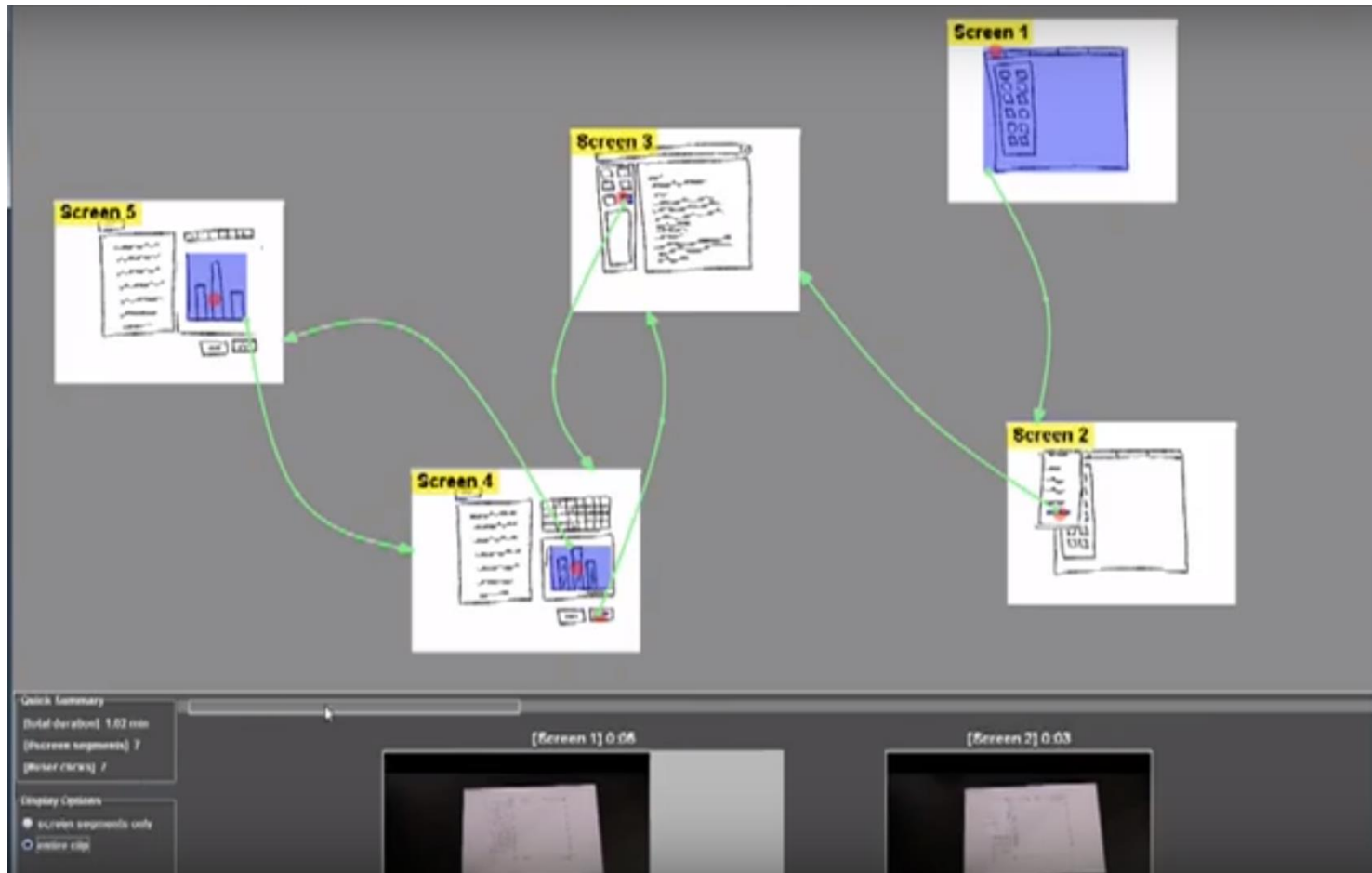
- Limitations
 - An additional person is required to conduct the test session
 - Paper prototypes are less suitable for visually-complex or highly-interactive interfaces
 - Requires some imagination on users part
 - forces you to stay in “early design” mode

Evaluation Example

- <https://www.youtube.com/watch?v=GrV2SZuRPv0>



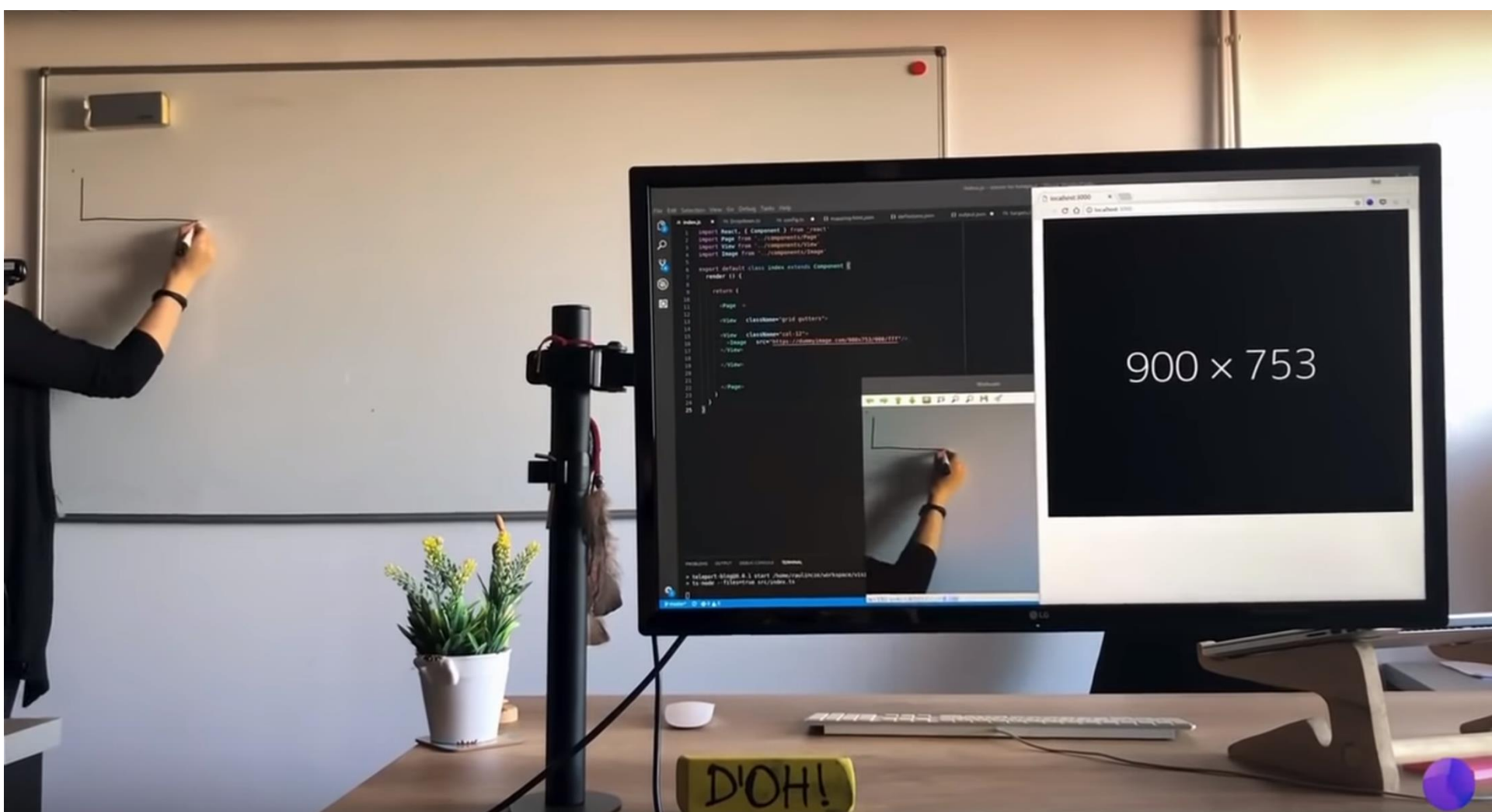
FrameWire: Automatically Extracting Interaction Logic from Paper Prototyping Tests



- <http://www.youtube.com/watch?v=q6pGIOhr3d0&feature=youtu.be>

Real-time Code Generation

- https://www.youtube.com/watch?v=_oet4GOzcRQ



Paper Prototyping Image Quality

- Rendering: How good should it look?
 - Good enough to elicit feedback about critical/risky design issues
- Creation speed is of the essence!
- **Modifications** - when?
 - While user is present, get immediate feedback.
 - Make changes without losing users' context.
- Coding effort in paper prototyping is always zero
 - No code to write or rewrite until design is stabilized

What you can learn from a paper prototype

- **Conceptual model**

- Do users understand it?

- **Functionality**

- Does it do what's needed? Missing features?

- **Navigation & task flow**

- Can users find their way around?
- Are information preconditions met?

- **Terminology**

- Do users understand labels?

- **Screen contents**

- What needs to go on the screen?

What you can't learn

- **Look:** color, font, whitespace, etc
- **Feel:** efficiency issues
- **Response time**
- Are **small changes** noticed?
 - Even the tiniest change to a paper prototype is clearly visible to user
- **Exploration vs. deliberation**
 - Users are more deliberate with a paper prototype; they don't explore or thrash as much

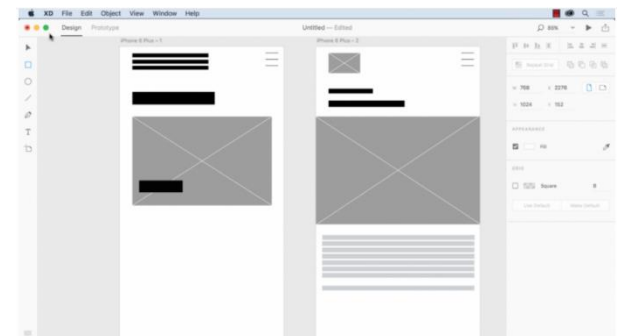
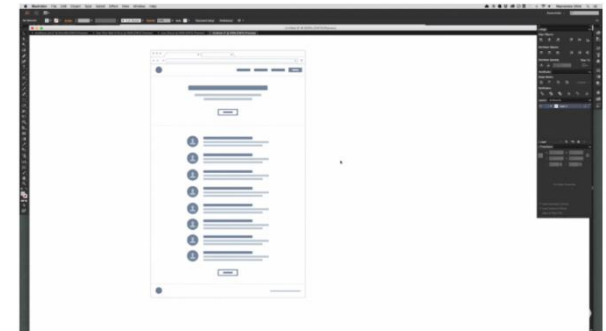
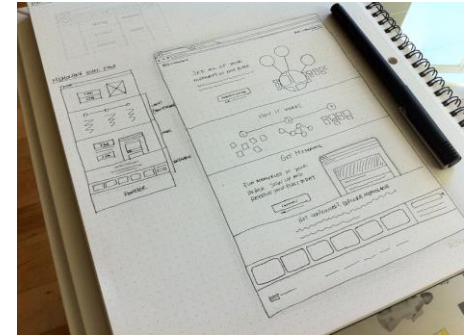
Wireframes

- A wireframe:
 - presents the information that will be displayed on the page
 - gives an outline of structure and layout of the page
 - conveys the overall direction and description of the user interface
- Note that wireframes are often low to medium fidelity.



Methods Of Wireframing

- Sketching: Hand-drawn sketches are quick to create and quick to iterate on
- Graphic design software: Adobe Photoshop or Illustrator.
- UX design software: Wireframe.cc, Balsamiq, Adobe Xd



Low-Fidelity Prototyping

- Pros

- Inexpensive: extremely low cost.
- Fast: quickly explore different ideas without too much effort
- Collaborative: people can be involved in the design process

- Cons

- Limited interactivity: Hard to convey complex animations or transitions
- Misleading:

Users often more inclined to comment on designs that appear less “finished”

Client (i.e., the person paying you) might be confused by the lack of programming